

PFAS in Miami Beach Drinking Water

Levels Are Dangerously High

Measure	PFOS (ppt)	PFOA (ppt)
Miami Beach 2023	38	30
EPA 2024 MCL (Goal: 0)	4	4
EPA 2022 Health Advisory	0.02	0.004

Miami Beach PFOS is 1,900× the EPA health advisory and 9.5× the MCL

Miami Beach PFOA is 7,500× the EPA health advisory and 7.5× the MCL

Sources: City of Miami Beach 2023 Water Quality Report; EPA PFAS Health Advisories (2022); EPA Final PFAS National Primary Drinking Water Regulation (2024)

What is PFAS? "Forever Chemicals"

PFAS = Per- and Polyfluoroalkyl Substances (a large class of chemicals)

PFOA and PFOS are the most common and dangerous PFAS in drinking water

Used in AFFF: Aqueous Film-Forming Foam (white firefighting foam used at airports and military facilities)

"Forever Chemicals": Do not degrade in environment or human body—
bio accumulate, persist

US EPA states PFAS can cause:

- **Reproductive effects such as decreased fertility or increased high blood pressure in pregnant women.**
- **Developmental effects or delays in children, including low birth weight, accelerated puberty, bone variations, or behavioral changes.**
- **Increased risk of cancers, including prostate, kidney, and testicular cancers.**

US EPA states PFAS harms children more

- Because children are still developing, they are more sensitive to the harmful effects
- They are exposed more than adults because children drink more water per pound of body weight which increases their exposure to PFAS.

SOURCE: <https://www.epa.gov/pfas/our-current-understanding-human-health-and-environmental-risks-pfas>

AFFF Foam Use at Miami International Airport

White firefighting foam deployed during aircraft emergencies, training burns, and crashes



Images: Miami International Airport firefighting foam deployment during aircraft emergencies (Getty Images, News sources)

PFAS at Miami International Airport

PFAS Detection:

242 ppt total PFAS detected in Grapeland Heights tap water (nearest MIA sampling location)

- ~30 different PFAS compounds found in Miami tap water, including PFOS, PFOA, PFPeA, PFHxS, PFHxA, PFBA, PFBS
- Highest concentrations near Miami International and Fort Lauderdale airports

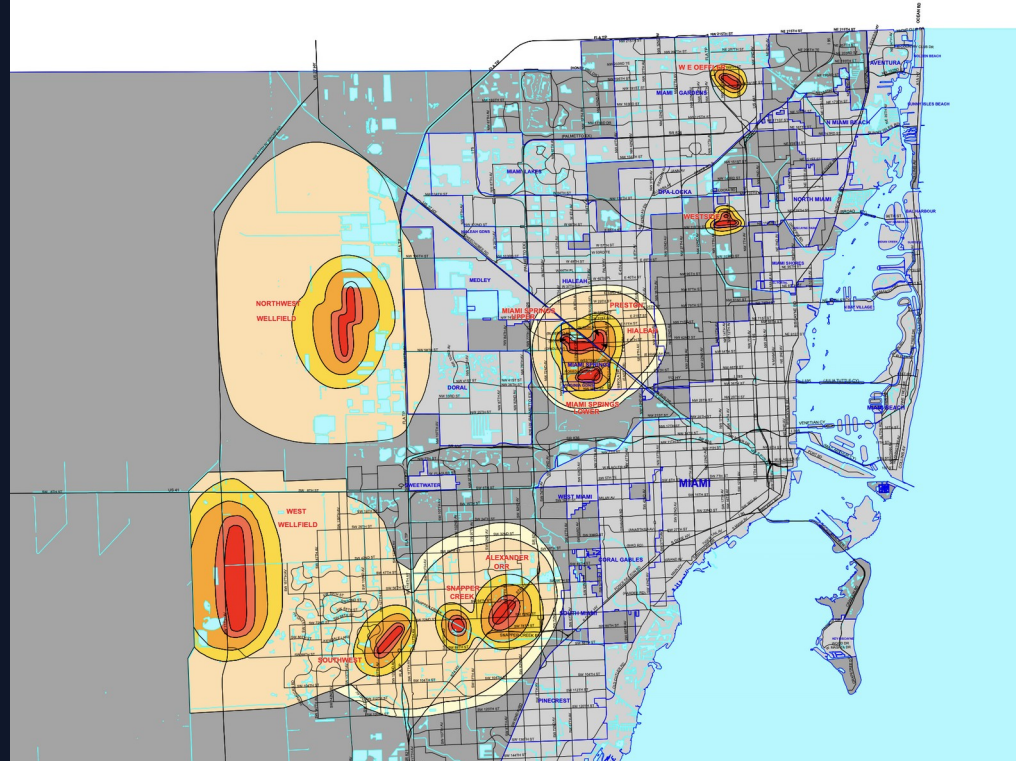
Miami-Dade County lawsuit vs 3M: Suing for \$200 million for PFAS contamination

Miami Dade County Attorney states: "the levels of PFAS in the County's drinking water supply relative to the projected state and federal MCL are high"

Miami Dade Wells

Miami-Dade County

Wellfield Protection Areas



FIU Research Papers

FIU Research Findings:

- **6:2 FTS fingerprint:** FIU detected 6:2 fluorotelomer (airport AFFF signature) dominating Miami River—directly flows from MIA to Biscayne Bay
- **Concentration gradient:** PFAS highest near airport, then canals (especially Miami River), decreasing into bay and offshore, but still elevated throughout bay and coastal waters
- **Marine contamination:** Dolphins, fish, and manatees in Biscayne Bay bioaccumulate PFAS at harmful levels
- PFAS chemicals spilling into water 24/7/365

<https://www.sciencedirect.com/science/article/abs/pii/S0048969723070419?via%3Dihub>

<https://pubmed.ncbi.nlm.nih.gov/35780268/>

Miami Dade County Treatment Timeline

Central PFAS treatment (RO/IX/GAC plant) not operational until 2029-2031

Pilots

Meanwhile: Children drinking 38 ppt PFOS / 30 ppt PFOA while county report calls it "excellent"

6+ year delay while PFAS accumulates in children, pregnant women, sensitive populations and environment

City of MB Water Quality Report is same as County Water Quality Report

What CMB Says (Direct Quote):

*"The Water Quality Report is designed to inform you about the **excellent water** the City of Miami Beach delivers to you every day."*

*"Our number one goal is to provide you and your family a **safe and dependable** supply of drinking water."*

*Report includes images of **smiling children drinking tap water** to convey safety and purity.*

The Reality (Same Report):

- PFOS: 38 ppt (vs 0.02 ppt advisory, 4 ppt MCL)
- PFOA: 30 ppt (vs 0.004 ppt advisory, 4 ppt MCL)
- EPA: cancer, immune suppression, and developmental effects in children

**Legal Liability, Health and Economic Risks Grow
Every Day**

Solution 1: Update Report & Educate Public

Immediate Actions:

Update MB Water Quality Report

- Use exact EPA language, remove excellent, safe, etc.
- Remove smiling child imagery from water safety marketing

Actively Promote Water Filtration

Reverse Osmosis: Removes 99%+ of PFAS and contaminants

Solution 2: Independent Testing & County Partnership

Three-Part Action:

1. Independent Testing

western wells?

2. Demand Seat at Table w/ County

3. Push County Action

- Accelerate
- Prioritize airport
AFFF source capping
- Promote plume
remediation between
airport and Biscayne
Bay

Solution 3: Decentralized RO Program (Fastest Path to Clean Water, treat at tap)

Why Reverse Osmosis?

Proven Technology: Removes 99%+ of PFAS, PFOA, PFOS, and all other contaminants

Already in Use: North Miami Beach, Hialeah, and other Florida utilities use RO successfully

Household & Municipal: Can be deployed at individual homes AND in schools, hospitals, facilities

Miami Beach Scale & Timeline

90,000 people, likely < 40,000 homes. Prioritize children, pregnant, sensitive populations, schools, day cares, etc. to start, get to everyone within 6 months.

Up and running within 90 days for guaranteed clean water.

Miami Beach has an opportunity to become a national model for decentralized, rapid-response PFAS remediation while protecting 90,000 residents. Public Works can lead this.